

Algebra II with Trigonometry

Chapter 6.5

Olympiad Solutions (GPT-5.4)

Saved tutoring response with MathJax rendering. Original PDF file:
[4533652_alg-2trig-h-chp-6-5-note.pdf](#)

Solutions

1. Evaluate exactly:

$$\arcsin\left(\frac{3}{5}\right) + \arccos\left(\frac{12}{13}\right) + \arctan\left(\frac{1}{3}\right).$$

Let

$$A = \arcsin\left(\frac{3}{5}\right), \quad B = \arccos\left(\frac{12}{13}\right), \quad C = \arctan\left(\frac{1}{3}\right).$$

Then all three angles are acute.

From the given values,

$$\begin{aligned} \sin A &= \frac{3}{5}, \quad \cos A = \frac{4}{5} \implies \tan A = \frac{3}{4}, \\ \cos B &= \frac{12}{13}, \quad \sin B = \frac{5}{13} \implies \tan B = \frac{5}{12}, \\ \tan C &= \frac{1}{3}. \end{aligned}$$

Now

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B} = \frac{\frac{3}{4} + \frac{5}{12}}{1 - \frac{3}{4} \cdot \frac{5}{12}} = \frac{\frac{7}{6}}{\frac{11}{16}} = \frac{56}{33}.$$

Then

$$\tan(A + B + C) = \frac{\frac{56}{33} + \frac{1}{3}}{1 - \frac{56}{33} \cdot \frac{1}{3}} = \frac{\frac{67}{33}}{\frac{43}{99}} = \frac{201}{43}.$$

This is not immediately recognizable, so a better approach is to combine B and C first:

$$\tan(B + C) = \frac{\frac{5}{12} + \frac{1}{3}}{1 - \frac{5}{12} \cdot \frac{1}{3}} = \frac{\frac{3}{4}}{\frac{31}{36}} = \frac{27}{31}.$$

Hence

$$\tan(A + B + C) = \frac{\frac{3}{4} + \frac{27}{31}}{1 - \frac{3}{4} \cdot \frac{27}{31}} = \frac{\frac{201}{124}}{\frac{43}{124}} = \frac{201}{43}.$$

So the sum is an acute angle with tangent $201/43$, but that still does not identify it.

Instead, compute sines and cosines directly for $B + C$:

$$\cos(B + C) = \cos B \cos C - \sin B \sin C = \frac{12}{13} \cdot \frac{3}{\sqrt{10}} - \frac{5}{13} \cdot \frac{1}{\sqrt{10}} = \frac{31}{13\sqrt{10}}$$

$$\sin(B + C) = \sin B \cos C + \cos B \sin C = \frac{5}{13} \cdot \frac{3}{\sqrt{10}} + \frac{12}{13} \cdot \frac{1}{\sqrt{10}} = \frac{27}{13\sqrt{10}}$$

Also

$$\sin A = \frac{3}{5}, \quad \cos A = \frac{4}{5}.$$

Thus

$$\sin(A + B + C) = \sin A \cos(B + C) + \cos A \sin(B + C) = \frac{3}{5} \cdot \frac{31}{13\sqrt{10}} + \frac{4}{5}$$

which is not special.

So let us instead test whether the angle is $\pi - \theta$ for some standard θ ; but since each term is acute and their approximate values are

$$\arcsin \frac{3}{5} \approx 36.87^\circ, \quad \arccos \frac{12}{13} \approx 22.62^\circ, \quad \arctan \frac{1}{3} \approx 18.43^\circ,$$

their sum is

$$36.87^\circ + 22.62^\circ + 18.43^\circ = 77.92^\circ.$$

Thus the exact value is not a standard rational multiple of π .

Therefore the exact simplified result is simply

$$\boxed{\arctan\left(\frac{201}{43}\right)}.$$

Since the sum lies in $(0, \pi/2)$, the arctangent gives the angle itself.

2. Compute the exact value of

$$\tan\left(2 \arcsin\left(\frac{3}{5}\right) - \arccos\left(\frac{5}{7}\right)\right).$$

Let

$$A = \arcsin\left(\frac{3}{5}\right), \quad B = \arccos\left(\frac{5}{7}\right).$$

Then

$$\sin A = \frac{3}{5}, \quad \cos A = \frac{4}{5}, \quad \tan A = \frac{3}{4}.$$

Hence

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A} = \frac{2 \cdot \frac{3}{4}}{1 - \frac{9}{16}} = \frac{\frac{3}{2}}{\frac{7}{16}} = \frac{24}{7}.$$

Also

$$\cos B = \frac{5}{7}, \quad \sin B = \frac{2\sqrt{6}}{7},$$

so

$$\tan B = \frac{2\sqrt{6}}{5}.$$

Now

$$\tan(2A - B) = \frac{\tan 2A - \tan B}{1 + \tan 2A \tan B} = \frac{\frac{24}{7} - \frac{2\sqrt{6}}{5}}{1 + \frac{24}{7} \cdot \frac{2\sqrt{6}}{5}}.$$

Put over common denominators:

$$= \frac{\frac{120 - 14\sqrt{6}}{35}}{\frac{35 + 48\sqrt{6}}{35}} = \frac{120 - 14\sqrt{6}}{35 + 48\sqrt{6}}.$$

Rationalizing,

$$\frac{120 - 14\sqrt{6}}{35 + 48\sqrt{6}} \cdot \frac{35 - 48\sqrt{6}}{35 - 48\sqrt{6}} = \frac{(120 - 14\sqrt{6})(35 - 48\sqrt{6})}{35^2 - (48\sqrt{6})^2}.$$

The denominator is

$$1225 - 13824 = -12599.$$

For the numerator,

$$(120)(35) - 120 \cdot 48\sqrt{6} - 14\sqrt{6} \cdot 35 + 14 \cdot 48 \cdot 6 = 4200 - 5760\sqrt{6} - 490\sqrt{6}$$

Thus

$$\tan(2A - B) = \frac{8232 - 6250\sqrt{6}}{-12599} = \frac{6250\sqrt{6} - 8232}{12599}.$$

So the exact value is

$$\frac{6250\sqrt{6} - 8232}{12599}.$$

3. Find the exact value of

$$\arcsin\left(\frac{4}{5}\right) - \arctan\left(\frac{4}{3}\right).$$

Let

$$A = \arcsin\left(\frac{4}{5}\right), \quad B = \arctan\left(\frac{4}{3}\right).$$

Both are acute.

From $\sin A = \frac{4}{5}$, we get

$$\cos A = \frac{3}{5}, \quad \tan A = \frac{4}{3}.$$

But $B = \arctan\left(\frac{4}{3}\right)$ is exactly the acute angle whose tangent is $\frac{4}{3}$.

Since $A, B \in (0, \pi/2)$ and $\tan A = \tan B$, we must have $A = B$. Therefore

$$\arcsin\left(\frac{4}{5}\right) - \arctan\left(\frac{4}{3}\right) = 0.$$

Hence

$$\boxed{0}.$$

4. Simplify to a single constant:

$$\arccos\left(\frac{1}{3}\right) + \arccos\left(-\frac{1}{3}\right) + \arcsin\left(\frac{5}{13}\right) + \arcsin\left(\frac{12}{13}\right).$$

Use the identity

$$\arccos(-x) = \pi - \arccos(x).$$

Therefore

$$\arccos\left(\frac{1}{3}\right) + \arccos\left(-\frac{1}{3}\right) = \arccos\left(\frac{1}{3}\right) + \pi - \arccos\left(\frac{1}{3}\right) = \pi.$$

Now let

$$A = \arcsin\left(\frac{5}{13}\right), \quad B = \arcsin\left(\frac{12}{13}\right).$$

Both are acute, and

$$\begin{aligned} \sin A &= \frac{5}{13}, & \cos A &= \frac{12}{13}, \\ \sin B &= \frac{12}{13}, & \cos B &= \frac{5}{13}. \end{aligned}$$

Thus

$$A + B = \frac{\pi}{2}.$$

So the whole expression equals

$$\pi + \frac{\pi}{2} = \frac{3\pi}{2}.$$

Hence

$$\boxed{\frac{3\pi}{2}}.$$

5. Let

$$x = \arctan 2 + \arctan 3 + \arctan 4.$$

Determine the exact value of $\tan x$.

First combine the first two terms:

$$\tan(\arctan 2 + \arctan 3) = \frac{2 + 3}{1 - 2 \cdot 3} = \frac{5}{-5} = -1.$$

Since $\arctan 2$ and $\arctan 3$ are both acute, their sum lies in $(\pi/2, \pi)$, so the angle is

$$\arctan 2 + \arctan 3 = \pi - \frac{\pi}{4} = \frac{3\pi}{4}.$$

Hence

$$x = \frac{3\pi}{4} + \arctan 4.$$

Therefore

$$\tan x = \frac{\tan \frac{3\pi}{4} + \tan(\arctan 4)}{1 - \tan \frac{3\pi}{4} \tan(\arctan 4)} = \frac{-1 + 4}{1 - (-1)(4)} = \frac{3}{5}.$$

So

$$\boxed{\frac{3}{5}}.$$

6. Find the exact value of

$$\cos\left(\arcsin\left(\frac{7}{25}\right) + 2\arctan\left(\frac{1}{3}\right)\right).$$

Let

$$A = \arcsin\left(\frac{7}{25}\right), \quad B = \arctan\left(\frac{1}{3}\right).$$

Then

$$\sin A = \frac{7}{25}, \quad \cos A = \frac{24}{25}.$$

Also, with $\tan B = \frac{1}{3}$,

$$\sin B = \frac{1}{\sqrt{10}}, \quad \cos B = \frac{3}{\sqrt{10}}.$$

So

$$\cos 2B = \cos^2 B - \sin^2 B = \frac{9}{10} - \frac{1}{10} = \frac{4}{5},$$

$$\sin 2B = 2 \sin B \cos B = 2 \cdot \frac{1}{\sqrt{10}} \cdot \frac{3}{\sqrt{10}} = \frac{3}{5}.$$

Now apply the addition formula:

$$\cos(A + 2B) = \cos A \cos 2B - \sin A \sin 2B = \frac{24}{25} \cdot \frac{4}{5} - \frac{7}{25} \cdot \frac{3}{5}.$$

Thus

$$\cos(A + 2B) = \frac{96}{125} - \frac{21}{125} = \frac{75}{125} = \frac{3}{5}.$$

Hence

$$\boxed{\frac{3}{5}}.$$